

ZSWIM8-6 旗舰方向 · 六类基金"即用型"完整草稿集

方向: Energy-stress-driven metabolic memory via ZSWIM8-dependent miRNA decay (AMPK 磷酸化 ZSWIM8-S608 → 加速代谢 miRNA 的 TDMD → T2D 中可逆的代谢记忆) 申请人: Xiaodong (Sheldon) Zou · Eric Lai Lab, MSKCC 生成日期: 2026-07-22

使用说明: 以下六份草稿已按各基金的真实章节结构与篇幅预先裁剪好,ZSWIM8-6 内容已填入。你只需把 ■FILL: ...■ 标记替换成真实数据/履历/预算即可投稿,无需再从母版剪裁。所有预期结果图/数字均标注 predicted/schematic。

目录

- 模板 A · NIH K99/R00 (博士后→独立过渡,唯一对非美籍开放的直通车)
- 模板 B · NIH NRSA F32 (需美籍/绿卡,重训练)
- 模板 C · NIH R01 (独立 PI 项目,结构模板)
- 模板 D · 海外优青 (正文可中文,重代表作+国家需求)
- 模板 E · 香港 RGC ECS (重可考核目标+36 月可行性)
- 模板 F · 新加坡 NRF Fellowship (重愿景+个人 track record)

模板 A · NIH K99/R00 (Pathway to Independence) — 完整草稿

目标 NOFO: PAR-25-135 (NCI) · 研究计划总篇幅 ≤12 页 (K99 导师期 6 页 + R00 独立期 6 页) · 正文英文 你的定位: 博士后≤4 年、不限国籍——这是你唯一能在博士后阶段直接申、且对非美籍开放的通道。

SPECIFIC AIMS (1 页 · 独立成页)

MicroRNAs, once regarded as constitutively stable, are actively destroyed on a physiological timescale by target-directed miRNA decay (TDMD), in which the ZSWIM8 ubiquitin ligase recognizes a highly complementary target engaged by an AGO2-miRNA complex and marks AGO2 for degradation. Although the ZSWIM8 TDMD machinery is now structurally defined, whether and how it is tuned by the cell's metabolic state is unknown. Whether the energy sensor AMPK—activated within minutes of nutrient stress—directly controls the rate of miRNA turnover has never been tested. This gap is pressing because type 2 diabetes (T2D) exhibits "metabolic memory": tissues retain a disordered metabolic program long after glycemic control is restored, yet no molecular mechanism explains this persistence.

Our long-term goal is to define the metabolic control of miRNA stability and exploit it therapeutically. Our central hypothesis is that energy stress, through AMPK, phosphorylates ZSWIM8 at S608, accelerating TDMD of specific metabolic miRNAs and thereby encoding a durable, reversible metabolic memory. This hypothesis is grounded in our transparent AMPK substrate-motif scan, which places ZSWIM8-S608 in a textbook AMPK recognition context (...KRR-S608...) within a disordered, kinase-accessible region.

Aim 1 — Test whether AMPK directly phosphorylates ZSWIM8 at S608. In vitro kinase assays with recombinant AMPK and ZSWIM8 fragments, phospho-specific detection, and S608A/S608D mutants will establish direct phosphorylation and its dependence on the predicted motif. Predicted outcome: S608 phosphorylation is AMP-dependent and abolished by S608A.

Aim 2 — Determine how S608 phosphorylation sets the rate of TDMD. Using time-resolved small-RNA sequencing after transcriptional shut-off in ZSWIM8-reconstituted cells (WT / S608A / S608D / KO), we will fit decay-rate constants for candidate miRNAs. Predicted ordering: S608D > WT > S608A > KO.

Aim 3 — Test whether AMPK-driven ZSWIM8 activity encodes metabolic memory. In a T2D pig model and in fibrosis organoids subjected to a defined nutrient-stress pulse, we will ask whether a transient stress produces a durable, S608-dependent shift in metabolic miRNA levels and phenotype. Predicted: transient stress → lasting, S608-dependent miRNA suppression.

Successful completion will establish the first metabolic input to the TDMD machinery, define a phospho-switch (ZSWIM8-S608) as a druggable node, and open metabolic control of miRNA stability as a new axis in diabetes and its complications.

A. CANDIDATE'S BACKGROUND (候选人背景 · 约 1 页)

I trained in biochemistry and molecular biology (PhD, 2022) with a focus on AMPK energy-stress signaling and small-RNA analysis. My seven first-author publications in RNA biology and metabolic disease establish my capacity to lead this interdisciplinary program. (FILL: 用 3–4 句点出你最相关的 2 篇代表作与它们如何为本项目铺路——例如小 RNA 测序体系的建立、AMPK 信号的功能研究。)

B. CAREER DEVELOPMENT / TRAINING PLAN (职业发展与训练计划 · 约 1.5 页)

Objective. To become an independent investigator studying the metabolic control of miRNA stability. Training goals (mentored K99 phase): (1) quantitative RNA-decay kinetics and time-resolved small-RNA-seq analysis; (2) in vivo large-animal (T2D pig) metabolic modeling; (3) structural/biochemical reconstitution of the ZSWIM8-AGO2 complex; (4) responsible conduct of research, rigor and reproducibility, and lab-management/mentoring coursework. During the mentored phase I will acquire advanced training in quantitative RNA-decay measurement and in vivo large-animal metabolic modeling—skills I will carry into my independent laboratory. (FILL:

插入一张训练时间表——每半年一个里程碑,列课程名、要掌握的技术、导师/合作者。评审最看重 mentored→independent 的逻辑连贯性。)

C. TRAINING ACTIVITIES & MENTORING TEAM (训练活动与导师团队 · 约 1 页)

My mentoring/collaborator team spans TDMD biochemistry, kinase signaling, and organoid disease modeling, covering every methodology in this proposal. (FILL: 列 3–4 位导师/顾问姓名与各自负责的模块 (TDMD 结构、激酶信号、类器官、统计)。) The institutional environment provides (FILL: 核心设施——测序中心、动物设施、生信支持。)

D. RESEARCH STRATEGY (研究策略 · 约 6 页,K99+R00 分段)

Significance

- Clinical burden. Metabolic memory drives the persistence of diabetic complications despite glycemic control; no therapy erases this memory. A metabolic lever on miRNA decay would be a mechanistically novel entry point.
- Mechanistic advance. Establishing AMPK as an upstream regulator of TDMD would convert miRNA turnover from a housekeeping process into a signal-responsive, druggable pathway.
- Conceptual reach. Because ZSWIM8 governs decay of many miRNAs, a metabolic input would have consequences across the miRNAome, not a single miRNA.
- Unique resources. Direct access to a T2D pig model, intestinal/cardiac organoids, a lab-native small-RNA sequencing pipeline, and base/prime editing positions us to test this axis in human-relevant, large-animal contexts other laboratories cannot easily replicate.

Innovation

- Conceptual: a paradigm shift—miRNA decay is not constitutive but is set by metabolic state through a defined kinase→ligase axis.
- New target: ZSWIM8-S608 as, to our knowledge, the first phosphosite proposed to couple energy sensing to TDMD.
- Methodological: an integrated pipeline—transparent AMPK substrate scorer + endogenous phospho-mimetic editing + single-miRNA decay kinetics—not previously applied to TDMD.
- Editing: base/prime editing installs precise endogenous S608A/S608D alleles, avoiding overexpression artifacts.

Approach — K99 mentored phase (在导师指导下完成的过渡性工作)

Aim 1 (Yr 1–2) . Rationale: our motif scan flags S608 (...KRR-S608...) in a disordered, accessible region. Design: recombinant AMPK + ZSWIM8-fragment kinase assays $\pm$ AMP; phospho-specific antibody/mass spectrometry; S608A and S608D point mutants. Readout: stoichiometry of S608 phosphorylation. Predicted result (schematic): AMP-dependent phosphorylation, abolished by S608A. Alternative: if S608 is insufficient, test the neighboring S609 and S1202 sites (also flagged by our scan) and screen AMPK-independent kinases. Pitfall & mitigation: S608D may not fully mimic phospho-S608; corroborate with in-vitro-phosphorylated protein and phospho-specific antibodies.

Approach — R00 independent phase (你独立实验室的招牌工作)

Aim 2 (Yr 3–4) . Design: reconstitute ZSWIM8-KO cells with WT / S608A / S608D; halt transcription; perform a time-course small-RNA-seq. Readout: per-miRNA decay-rate constant k . Predicted result (schematic): S608D > WT > S608A > KO. Statistics: powered to detect a 30% difference in k at 80% power ($\alpha=0.05$), ≥ 3 biological replicates. Pitfall & mitigation: to separate global from selective effects, normalize to TDMD-insensitive miRNAs.

Aim 3 (Yr 4–5) . Design: (i) T2D pig tissue under a controlled metabolic challenge; (ii) intestinal/cardiac fibrosis organoids given a defined nutrient-stress pulse and tracked for 7–14 days, with endogenous S608A/S608D installed by prime editing. Readout: durable change in metabolic miRNA levels and phenotype after stress withdrawal. Predicted result (schematic): transient stress $\rightarrow$ lasting, S608-dependent suppression (metabolic memory). Alternative: if organoids lack durability, extend the pulse or apply repeated pulses.

Rigor & reproducibility: both sexes will be studied and sex included as a biological variable; all antibodies, cell lines and ZSWIM8 constructs authenticated and reported per RRID standards; a go/no-go milestone requires Aim 1 to confirm S608 phosphorylation before Aim 2 kinetics proceed. The independent phase establishes a program on the metabolic control of miRNA stability that is distinct from my mentor's focus on TDMD structure, giving me clear intellectual ownership.

(关键裁剪点:K99 段只放 Aim 1(打基础、需要训练的部分);R00 段放 Aim 2–3(你带走独立做的部分)。两段的工作必须明显能拆开——这是 K99/R00 评审的核心。)

投前检查 (K99/R00 专用)

- [] Aim 1 归 K99、Aim 2-3 归 R00,且彼此独立
- [] 训练计划有具体课程/技能/半年里程碑表
- [] 导师团队覆盖每一种方法学
- [] SABV(两性)+ RRID 试剂认证已写入 Approach
- [] 预实验图凡有真实数据替换 predicted/schematic
- [] 篇幅:研究计划 ≤12 页;字体/页边距按 SF424 Guide

模板 B · NIH NRSA F32 (博后个人 Fellowship) — 完整草稿

研究计划篇幅 ≤6 页 · 重"训练"轻"产出" · 正文英文 □□ 资格: F32 通常要求美籍/绿卡——动笔前先核对该 NOFO 的 citizenship 条款,不符则改投 K99。

SPECIFIC AIMS (1 页)

MicroRNAs, once regarded as constitutively stable, are actively destroyed on a physiological timescale by target-directed miRNA decay (TDMD), in which the ZSWIM8 ubiquitin ligase recognizes a highly complementary target engaged by an AGO2-miRNA complex and marks AGO2 for degradation. Although the ZSWIM8 TDMD machinery is now structurally defined, whether and how it is tuned by the cell's metabolic state is unknown. Whether the energy sensor AMPK—activated within minutes of nutrient stress—directly controls the rate of miRNA turnover has never been tested. This gap is pressing because type 2 diabetes (T2D) exhibits "metabolic memory": tissues retain a disordered metabolic program long after glycemic control is restored, yet no molecular mechanism explains this persistence.

Our long-term goal is to define the metabolic control of miRNA stability and exploit it therapeutically. Our central hypothesis is that energy stress, through AMPK, phosphorylates ZSWIM8 at S608, accelerating TDMD of specific metabolic miRNAs and thereby encoding a durable, reversible metabolic memory. This hypothesis is grounded in our transparent AMPK substrate-motif scan, which places ZSWIM8-S608 in a textbook AMPK recognition context (...KRR-S608...) within a disordered, kinase-accessible region.

Aim 1 — Test whether AMPK directly phosphorylates ZSWIM8 at S608. In vitro kinase assays with recombinant AMPK and ZSWIM8 fragments, phospho-specific detection, and S608A/S608D mutants will establish direct phosphorylation and its dependence on the predicted motif. Predicted outcome: S608 phosphorylation is AMP-dependent and abolished by S608A.

Aim 2 — Determine how S608 phosphorylation sets the rate of TDMD. Using time-resolved small-RNA sequencing after transcriptional shut-off in ZSWIM8-reconstituted cells (WT / S608A / S608D / KO), we will fit decay-rate constants for candidate miRNAs. Predicted ordering: S608D > WT > S608A > KO.

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Successful completion will establish the first metabolic input to the TDMD machinery, define a phospho-switch (ZSWIM8-S608) as a druggable node, and open metabolic control of miRNA stability as a new axis in diabetes and its complications.

(F32 裁剪:通常只主打 Aim 1-2(可在训练期完成的部分),Aim 3 作为"future direction"一句带过,避免过度承诺。)

RESEARCH TRAINING PLAN (研究训练计划 · 约 3-4 页)

Background & Significance (压缩至半页)

Clinical burden. Metabolic memory drives the persistence of diabetic complications despite glycemic control; no therapy erases this memory. A metabolic lever on miRNA decay would be a mechanistically novel entry point. Mechanistic advance. Establishing AMPK as an upstream regulator of TDMD would convert miRNA turnover from a housekeeping process into a signal-responsive, druggable pathway.

Approach

Aim 1. Rationale: our motif scan flags S608 (...KRR-S608...) in a disordered, accessible region. Design: recombinant AMPK + ZSWIM8-fragment kinase assays ± AMP; phospho-specific antibody/mass spectrometry; S608A and S608D point mutants. Readout: stoichiometry of S608 phosphorylation. Predicted result (schematic): AMP-dependent phosphorylation, abolished by S608A. Alternative: if S608 is insufficient, test the neighboring S609 and S1202 sites (also flagged by our scan) and screen AMPK-independent kinases. Pitfall & mitigation: S608D may not fully mimic phospho-S608; corroborate with in-vitro-phosphorylated protein and phospho-specific antibodies.

Aim 2. Design: reconstitute ZSWIM8-KO cells with WT / S608A / S608D; halt transcription; perform a time-course small-RNA-seq. Readout: per-miRNA decay-rate constant k . Predicted result (schematic): S608D > WT > S608A > KO. Statistics: powered to detect a 30% difference in k at 80% power ($\alpha=0.05$), ≥ 3 biological replicates. Pitfall & mitigation: to separate global from selective effects, normalize to TDMD-insensitive miRNAs. Rigor & reproducibility: both sexes will be studied and sex included as a biological variable; all antibodies, cell lines and ZSWIM8 constructs authenticated and reported per RRID standards; a go/no-go milestone requires Aim 1 to confirm S608 phosphorylation before Aim 2 kinetics proceed.

TRAINING GOALS & OBJECTIVES (训练目标——F32 的核心,约 1 页)

What I will learn that I cannot do now: (1) quantitative miRNA-decay kinetics; (2) time-resolved small-RNA-seq experimental design and analysis; (3) rigorous biochemical reconstitution. How this training advances my path to independence: During the mentored phase I will acquire advanced training in quantitative RNA-decay measurement and in vivo large-animal metabolic modeling—skills I will carry into my independent laboratory. (FILL: 把每个训练目标对应到一个具体的实验/课程,并说明由团队里谁指导。F32 评审给分主要看"训练价值",不是"科学产出量"。)

SPONSOR(S) & TRAINING ENVIRONMENT (导师与训练环境 · 约 1 页)

My mentoring/collaborator team spans TDMD biochemistry, kinase signaling, and organoid disease modeling, covering every methodology in this proposal. (FILL: 导师的经费与既往培养博士后进入独立岗位的记录;机构核心设施。)

RESPONSIBLE CONDUCT OF RESEARCH (RCR · 半页,必填)

(FILL: 参加的 RCR 课程/研讨、频率、形式——F32 明确要求单列此节。)

投前检查 (F32 专用)

- ☐ 先确认 citizenship 资格
 - ☐ 主打 Aim 1-2; Aim 3 仅作展望
 - ☐ "训练目标"节最详实,每项对应实验+导师
 - ☐ RCR 单独成节
 - ☐ SABV + RRID 已写入
 - ☐ 研究计划 ≤ 6 页
-

模板 C · NIH R01 (独立研究项目) — 完整草稿

研究计划篇幅 ≤ 12 页 · 不含 Career 节 · 正文英文 你的定位: 现阶段你还不是独立 PI——把这份当"结构模板"学习, K99→R00→R01 同一逻辑放大;拿到独立岗位后即可直接用。

SPECIFIC AIMS (1 页)

MicroRNAs, once regarded as constitutively stable, are actively destroyed on a physiological timescale by target-directed miRNA decay (TDMD), in which the ZSWIM8 ubiquitin ligase recognizes a highly complementary target engaged by an AGO2-miRNA complex and marks AGO2 for degradation. Although the ZSWIM8 TDMD machinery is now structurally defined, whether and how it is tuned by the cell's metabolic state is unknown. Whether the energy sensor AMPK—activated within minutes of nutrient stress—directly controls the rate of miRNA turnover has never been tested. This gap is pressing because type 2 diabetes (T2D) exhibits "metabolic memory": tissues retain a disordered metabolic program long after glycemic control is restored, yet no molecular mechanism explains this persistence.

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Successful completion will establish the first metabolic input to the TDMD machinery, define a phospho-switch (ZSWIM8-S608) as a druggable node, and open metabolic control of miRNA stability as a new axis in diabetes and its complications.

RESEARCH STRATEGY (研究策略 · 约 11 页)

Significance (约 2 页)

- Clinical burden. Metabolic memory drives the persistence of diabetic complications despite glycemic control; no therapy erases this memory. A metabolic lever on miRNA decay would be a mechanistically novel entry point.
- Mechanistic advance. Establishing AMPK as an upstream regulator of TDMD would convert miRNA turnover from a housekeeping process into a signal-responsive, druggable pathway.
- Conceptual reach. Because ZSWIM8 governs decay of many miRNAs, a metabolic input would have consequences across the miRNAome, not a single miRNA.
- Unique resources. Direct access to a T2D pig model, intestinal/cardiac organoids, a lab-native small-RNA sequencing pipeline, and base/prime editing positions us to test this axis in human-relevant, large-animal contexts other laboratories cannot easily replicate. (FILL: R01 的 Significance 要比 fellowship 更深——补一段该领域现状综述 + 你的项目如何填补, 引 8–12 篇文献。)

Innovation (约 1 页)

- Conceptual: a paradigm shift—miRNA decay is not constitutive but is set by metabolic state through a defined kinase→ligase axis.
- New target: ZSWIM8-S608 as, to our knowledge, the first phosphosite proposed to couple energy sensing to TDMD.
- Methodological: an integrated pipeline—transparent AMPK substrate scorer + endogenous phospho-mimetic editing + single-miRNA decay kinetics—not previously applied to TDMD.
- Editing: base/prime editing installs precise endogenous S608A/S608D alleles, avoiding overexpression artifacts.

Approach (约 7–8 页, 预实验要充分)

Preliminary Data. (FILL: R01 最看重预实验。这里放你的 AMPK 底物 PSSM 打分结果、S608 基序分析图, 以及任何已有的 ZSWIM8/TDMD 数据。凡 predicted/schematic 的必须标注; R01 评审对“纯预期”的容忍度低于 fellowship, 尽量补真实数据。)

Aim 1. Rationale: our motif scan flags S608 (...KRR-S608...) in a disordered, accessible region. Design: recombinant AMPK + ZSWIM8-fragment kinase assays ± AMP; phospho-specific antibody/mass spectrometry; S608A and S608D point mutants. Readout: stoichiometry of S608 phosphorylation. Predicted result (schematic): AMP-dependent phosphorylation, abolished by S608A. Alternative: if S608 is insufficient, test the neighboring S609 and S1202 sites (also flagged by our scan) and screen AMPK-independent kinases. Pitfall & mitigation: S608D may not fully mimic phospho-S608; corroborate with in-vitro-phosphorylated protein and phospho-specific antibodies.

Aim 2. Design: reconstitute ZSWIM8-KO cells with WT / S608A / S608D; halt transcription; perform a time-course small-RNA-seq. Readout: per-miRNA decay-rate constant k . Predicted result (schematic): S608D > WT > S608A > KO. Statistics: powered to detect a 30% difference in k at 80% power ($\alpha=0.05$), ≥3 biological replicates. Pitfall & mitigation: to separate global from selective effects, normalize to TDMD-insensitive miRNAs.

Aim 3. Design: (i) T2D pig tissue under a controlled metabolic challenge; (ii) intestinal/cardiac fibrosis organoids given a defined nutrient-stress pulse and tracked for 7–14 days, with endogenous S608A/S608D installed by prime editing. Readout: durable change in metabolic miRNA levels and phenotype after withdrawal. Predicted result (schematic): transient stress → lasting, S608-dependent suppression (metabolic memory). Alternative: if organoids lack durability, extend the pulse or apply repeated pulses.

Rigor & reproducibility: both sexes will be studied and sex included as a biological variable; all antibodies, cell lines and ZSWIM8 constructs authenticated and reported per RRID standards; a go/no-go milestone requires Aim 1 to confirm S608 phosphorylation before Aim 2 kinetics proceed. (FILL: 每个 Aim 末尾加“Expected Outcomes / Interpretation / Potential Pitfalls & Alternative Approaches”三小段——R01 评审逐 Aim 找这三样。)

投前检查 (R01 专用)

- [] 需独立 PI 身份(现阶段作模板学习)
- [] Significance 有文献综述 + 更深论证
- [] Preliminary Data 单列, 尽量真实
- [] 每 Aim 有 Expected/Pitfalls/Alternatives 三段
- [] SABV + RRID + Authentication 全写
- [] 研究计划 ≤12 页

模板 D · 国家自然科学基金 海外优青 — 完整草稿 (正文可中文)

结构: 科学问题 → 研究基础 → 研究方案 → 代表作证据链 → 国家需求与回国工作设想 优势: 你的 7 篇一作 + 生化 PhD + 独特资源正好命中"代表作"与"研究基础"两栏。

一、拟解决的关键科学问题

能量应激如何被"写入"细胞的 miRNA 稳态并形成可逆的代谢记忆?我们提出:AMPK 直接磷酸化 TDMD 关键接头蛋白 ZSWIM8(S608),加速一批代谢相关 miRNA 的靶标导向降解(TDMD),从而把一次能量应激转化为持续、可逆的转录后记忆。 这一"激酶-TDMD"轴此前无人报道。

二、研究背景与研究基础

Clinical burden. Metabolic memory drives the persistence of diabetic complications despite glycemic control; no therapy erases this memory. A metabolic lever on miRNA decay would be a mechanistically novel entry point. Mechanistic advance. Establishing AMPK as an upstream regulator of TDMD would convert miRNA turnover from a housekeeping process into a signal-responsive, druggable pathway. Conceptual reach. Because ZSWIM8 governs decay of many miRNAs, a metabolic input would have consequences across the miRNAome, not a single miRNA.

申请人研究基础: (FILL: 用你的 7 篇一作串成证据链——代谢/乳酸/胆固醇/AMPK 方向的既往工作如何为本项目铺路;T2D 猪模型、类器官、碱基/先导编辑等独特资源;吉林大学生化博士训练。每篇代表作一句话点明与本课题的逻辑关系。)

三、研究方案 (对应三大目标)

目标一(机制): Rationale: our motif scan flags S608 (...KRR-S608...) in a disordered, accessible region. Design: recombinant AMPK + ZSWIM8-fragment kinase assays ± AMP; phospho-specific antibody/mass spectrometry; S608A and S608D point mutants. Readout: stoichiometry of S608 phosphorylation. Predicted result (schematic): AMP-dependent phosphorylation, abolished by S608A. Alternative: if S608 is insufficient, test the neighboring S609 and S1202 sites (also flagged by our scan) and screen AMPK-independent kinases. Pitfall & mitigation: S608D may not fully mimic phospho-S608; corroborate with in-vitro-phosphorylated protein and phospho-specific antibodies.

目标二(功能): Design: reconstitute ZSWIM8-KO cells with WT / S608A / S608D; halt transcription; perform a time-course small-RNA-seq. Readout: per-miRNA decay-rate constant k. Predicted result (schematic): S608D > WT > S608A > KO. Statistics: powered to detect a 30% difference in k at 80% power ($\alpha=0.05$), ≥3 biological replicates. Pitfall & mitigation: to separate global from selective effects, normalize to TDMD-insensitive miRNAs.

目标三(转化): Design: (i) T2D pig tissue under a controlled metabolic challenge; (ii) intestinal/cardiac fibrosis organoids given a defined nutrient-stress pulse and tracked for 7–14 days, with endogenous S608A/S608D installed by prime editing. Readout: durable change in metabolic miRNA levels and phenotype after stress withdrawal. Predicted result (schematic): transient stress → lasting, S608-dependent suppression (metabolic memory). Alternative: if organoids lack durability, extend the pulse or apply repeated pulses.

Rigor & reproducibility: both sexes will be studied and sex included as a biological variable; all antibodies, cell lines and ZSWIM8 constructs authenticated and reported per RRID standards; a go/no-go milestone requires Aim 1 to confirm S608 phosphorylation before Aim 2 kinetics proceed.

四、代表性成果 (代表作,≤5 篇)

(FILL: 列 5 篇一作,每篇标注期刊/年份/影响因子/你的贡献/被引;优先与本课题主线(代谢×miRNA)相关者。)

五、对国家重大需求的意义与回国工作设想

代谢性疾病(T2D 等)是我国重大慢病负担;阐明可逆代谢记忆的分子开关,为"停药后复发"提供全新干预节点。(FILL: 拟依托单位、拟组建团队规模、5 年目标、与国内已有平台的衔接。)

投前检查 (海外优青专用)

- [] 科学问题一句话可复述
 - [] 研究基础与 7 篇代表作形成证据链
 - [] 代表作≤5 篇,贡献与相关性写清
 - [] "国家需求 + 回国设想"单列且具体
 - [] 年龄/学位/海外经历资格自查
-

模板 E · 香港 RGC Early Career Scheme (ECS) — 完整草稿 (正文英文)

结构: Objectives → Background of Research → Research Plan & Methodology → Impact & Significance 定位: 面向刚入职香港高校的 AP;项目规模小于 R01,重"可行性 + 明确产出"。

OBJECTIVES (明确、可考核的目标列表)

- 1. Determine whether AMPK directly phosphorylates ZSWIM8 at S608 and define the kinase–substrate relationship.
- 2. Establish whether S608 phosphorylation accelerates TDMD of a defined set of metabolic miRNAs (keystone: miR-29).
- 3. Test whether this axis encodes a reversible "metabolic memory" in a disease-relevant model.

BACKGROUND OF RESEARCH

Clinical burden. Metabolic memory drives the persistence of diabetic complications despite glycemic control; no therapy erases this memory. A metabolic lever on miRNA decay would be a mechanistically novel entry point. Mechanistic advance. Establishing AMPK as an upstream regulator of TDMD would convert miRNA turnover from a housekeeping process into a signal-responsive, druggable pathway. Unique resources. Direct access to a T2D pig model, intestinal/cardiac organoids, a lab-native small-RNA sequencing pipeline, and base/prime editing positions us to test this axis in human-relevant, large-animal contexts other laboratories cannot easily replicate. (FILL: ECS 篇幅有限,背景 ≤1.5 页,聚焦"gap → 你的假说"。)

RESEARCH PLAN & METHODOLOGY

Objective 1. Rationale: our motif scan flags S608 (...KRR-S608...) in a disordered, accessible region. Design: recombinant AMPK + ZSWIM8-fragment kinase assays ± AMP; phospho-specific antibody/mass spectrometry; S608A and S608D point mutants. Readout: stoichiometry of S608 phosphorylation. Predicted result (schematic): AMP-dependent phosphorylation, abolished by S608A. Alternative: if S608 is insufficient, test the neighboring S609 and S1202 sites (also flagged by our scan) and screen AMPK-independent kinases. Pitfall & mitigation: S608D may not fully mimic phospho-S608; corroborate with in-vitro-phosphorylated protein and phospho-specific antibodies.

Objective 2. Design: reconstitute ZSWIM8-KO cells with WT / S608A / S608D; halt transcription; perform a time-course small-RNA-seq. Readout: per-miRNA decay-rate constant k. Predicted result (schematic): S608D > WT > S608A > KO. Statistics: powered to detect a 30% difference in k at 80% power (α=0.05), ≥3 biological replicates. Pitfall & mitigation: to separate global from selective effects, normalize to TDMD-insensitive miRNAs.

Objective 3. Design: (i) T2D pig tissue under a controlled metabolic challenge; (ii) intestinal/cardiac fibrosis organoids given a defined nutrient-stress pulse and tracked for 7–14 days, with endogenous S608A/S608D installed by prime editing. Readout: durable change in metabolic miRNA levels and phenotype after stress withdrawal. Predicted result (schematic): transient stress → lasting, S608-dependent suppression (metabolic memory). Alternative: if organoids lack durability, extend the pulse or apply repeated pulses.

Rigor & reproducibility: both sexes will be studied and sex included as a biological variable; all antibodies, cell lines and ZSWIM8 constructs authenticated and reported per RRID standards; a go/no-go milestone requires Aim 1 to confirm S608 phosphorylation before Aim 2 kinetics proceed. (FILL: ECS 评审很看"36 个月内可完成"——给一个季度粒度的时间线甘特图,并写明香港本地的核心设施/合作者。)

IMPACT & SIGNIFICANCE

Conceptual: a paradigm shift—miRNA decay is not constitutive but is set by metabolic state through a defined kinase→ligase axis. Editing: base/prime editing installs precise endogenous S608A/S608D alleles, avoiding overexpression artifacts. (FILL: 说明对香港/大湾区生物医药布局的贡献,以及培养研究生的计划。)

投前检查 (RGC ECS 专用)

- [] Objectives 可考核、编号清晰
- [] 背景 ≤1.5 页,直指 gap
- [] 36 个月甘特图 + 本地设施
- [] Impact 关联大湾区
- [] 需已获香港高校 AP offer

模板 F · 新加坡 NRF Fellowship — 完整草稿 (正文英文)

RESEARCH VISION (研究愿景——NRF 的灵魂,写大格局)

I will establish that cellular energy state is transduced into durable, reversible programs of miRNA turnover — a new layer of metabolic memory — and build an independent program mapping the "kinase–TDMD" interface across metabolic disease. Successful completion will establish the first metabolic input to the TDMD machinery, define a phospho-switch (ZSWIM8-S608) as a druggable node, and open metabolic control of miRNA stability as a new axis in diabetes and its complications. (FILL: NRF 要 10 年尺度的愿景,不只这个项目;写清你要开创的"子领域"。)

TRACK RECORD (个人履历与代表成果)

(FILL: 7 篇一作 + 生化 PhD(吉林大学)+ 代谢/AMPK/编辑技术专长 + 独特模型资源。NRF 极看重 PI 个人潜力,用量化指标(被引、期刊、独立性证据)。)

RESEARCH PLAN

Aim 1. Rationale: our motif scan flags S608 (...KRR-S608...) in a disordered, accessible region. Design: recombinant AMPK + ZSWIM8-fragment kinase assays ± AMP; phospho-specific antibody/mass spectrometry; S608A and S608D point mutants. Readout: stoichiometry of S608 phosphorylation. Predicted result (schematic): AMP-dependent phosphorylation, abolished by S608A. Alternative: if S608 is insufficient, test the neighboring S609 and S1202 sites (also flagged by our scan) and screen AMPK-independent kinases. Pitfall & mitigation: S608D may not fully mimic phospho-S608; corroborate with in-vitro-phosphorylated protein and phospho-specific antibodies.

Aim 2. Design: reconstitute ZSWIM8-KO cells with WT / S608A / S608D; halt transcription; perform a time-course small-RNA-seq. Readout: per-miRNA decay-rate constant k. Predicted result (schematic): S608D > WT > S608A > KO. Statistics: powered to detect a 30% difference in k at 80% power ($\alpha=0.05$), ≥3 biological replicates. Pitfall & mitigation: to separate global from selective effects, normalize to TDMD-insensitive miRNAs.

Aim 3. Design: (i) T2D pig tissue under a controlled metabolic challenge; (ii) intestinal/cardiac fibrosis organoids given a defined nutrient-stress pulse and tracked for 7–14 days, with endogenous S608A/S608D installed by prime editing. Readout: durable change in metabolic miRNA levels and phenotype after stress withdrawal. Predicted result (schematic): transient stress → lasting, S608-dependent suppression (metabolic memory). Alternative: if organoids lack durability, extend the pulse or apply repeated pulses.

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BUDGET & HOST INSTITUTION

(FILL: 5 年预算框架(人员/设备/耗材/动物)+ 拟依托的新加坡机构(如 A*STAR/NUS/NTU)及其平台契合度 + 招聘计划。)

投前检查 (NRF 专用)

- [] 愿景有 10 年尺度的雄心
- [] Track record 量化、突出独立性
- [] 5 年预算 + 依托机构契合度
- [] 需新加坡机构 host 支持